

Political distrust, armed conflict, and body reclamation drive Ebola transmission dynamics in eastern Democratic Republic of Congo: a coupled epidemic and opinion dynamics model (SEIHRF-OD)

Selain K. Kasereka, Kyandoghere Kyamakya, Jean-Jacques T. Muyembe

Institute of Smart Systems Technologies, University of Klagenfurt; Institut National de Recherche Biomédicale (INRB) / One Health Institute for Africa (INOHA), Kinshasa, Democratic Republic of Congo;

Correspondence: kyandoghere.kyamakya@aau.at

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Summary

Background. Standard compartmental models of Ebola virus disease assume behavioral homogeneity within the exposed population. The 2026 Bundibugyo ebolavirus (BDBV) outbreak in the Democratic Republic of Congo (DRC) violates this assumption: a substantial fraction of affected communities in eastern conflict-affected provinces actively deny that the disease is real, refuse hospitalisation, and reclaim the bodies of deceased relatives for traditional burial, a practice associated with catastrophic secondary transmission. No existing model captures this coupled epidemic and behavioral dynamic.

Methods. We developed the SEIHRF-OD model, a stratified ordinary-differential-equation (ODE) system coupling a dual-population epidemic layer (Believers B ; Skeptics N) with an opinion-dynamics layer $\phi(t)$. Skeptics contribute a novel compartment F_R (reclaimed bodies) with transmission coefficient β_{FR} that substantially exceeds community and hospital rates. Opinion evolves under social contagion (α), visible deaths (β_D), public-health communication (γ_{comm}), and armed-conflict intensity $C(t)$. We calibrate the model on daily INSP situation-report data and WHO weekly case counts (INRB-UMIE/Ebola_DRC_2026, build 235a3c3) using Bayesian Markov-chain Monte Carlo (MCMC) inference, derive the

basic reproduction number $\mathcal{R}_0$ analytically via the Next Generation Matrix, and simulate three counterfactual interventions.

Findings. $\mathcal{R}_0$ decomposes into a believer component $\mathcal{R}_0^B$ and a skeptic component $\mathcal{R}_0^N$, the latter amplified by a β_{FR} -dependent burial term absent from all prior Ebola models. Across plausible parameter ranges ($\phi_0 \in [0.20, 0.55]$, $\beta_{FR}/\beta_I \in [3, 7]$), the skeptic sub-population drives $\mathcal{R}_0^N$ 1.8 to $2.9\times$ above $\mathcal{R}_0^B$; the population-weighted $\mathcal{R}_0 \approx 2.4$, a 33 % overestimate relative to a homogeneous model fitted to the same data. Counterfactual simulations indicate that doubling communication effectiveness from day 14 would avert 47 % (95 % CrI: 27 to 56 %) of cumulative deaths at 90 days, and eliminating body reclamation would avert 38 % (20 to 57 %).

Interpretation. Ignoring belief heterogeneity causes systematic underestimation of transmission potential and misdirects intervention targets. Targeted communication that reduces $\phi(t)$ is as impactful as direct transmission-blocking measures. Conflict intensity $C(t)$ acts as a structural amplifier of skepticism and must be incorporated in outbreak response planning for conflict-affected settings.

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Research in context

Evidence before this study. We searched PubMed and medRxiv on 1 May 2026 using the terms Ebola AND (mathematical model OR compartmental model) AND (behavior OR trust OR belief) with no date restriction. The seminal Legrand et al. [1] SEIHDR model and its extensions [2, 3] stratify transmission by setting but assume a homogeneous, fully compliant population. Funk and colleagues introduced awareness-driven behavior into disease transmission [4, 5], yet did not integrate this with Ebola-specific transmission pathways or empirical conflict data. No prior study couples explicit opinion dynamics to an Ebola compartmental model or introduces a dedicated reclaimed-body compartment.

Added value of this study. This is the first Ebola model to: (i) stratify the population by

disease belief; (ii) introduce a reclaimed-body compartment F_R with an independently estimable transmission coefficient β_{FR} ; (iii) couple epidemic dynamics bidirectionally to a time-varying opinion process modulated by conflict intensity; and (iv) calibrate all components simultaneously on the highest-resolution public Ebola dataset currently available: 519-zone daily INSP sitreps plus WHO weekly data.

Implications of all available evidence. Standard SEIHFR models fitted to the 2026 DRC data would underestimate $\mathcal{R}_0$ and misjudge the relative impact of interventions. Policymakers should treat skepticism-driven body reclamation as a structural driver of transmission rather than a peripheral behavioral complication.

Introduction

Ebola virus disease (EVD) caused by the Bundibugyo ebolavirus (BDBV) was first identified in Uganda in 2007 and is associated with a case-fatality ratio of 25 to 40 % [6]. The 2026 outbreak in the DRC, declared on 15 May 2026 [20] and designated a PHEIC by WHO on 16 May 2026 [21], is the third BDBV outbreak on record and the first to emerge in an active conflict zone, affecting Ituri, North Kivu, and South Kivu provinces where armed groups and sustained military operations have disrupted health infrastructure for over a decade [11].

Mathematical models have been central to outbreak response since Legrand et al. [1] formalised the role of funeral transmission; their SEIHFR framework was extended to incorporate spatial spread [10], intervention timing [9], and real-time $\mathcal{R}_t$ estimation [8]. All existing models, however, share a critical assumption: that the population follows public-health guidance uniformly.

This assumption is violated in the 2026 DRC context. Field data collected by the Institut National de Santé Publique (INSP) across seven situation reports (SitReps 001 to 007) document widespread community rejection of the EVD diagnosis, rooted in political distrust associated with the ongoing conflict. Affected families have on multiple occa-

sions physically reclaimed the bodies of deceased relatives from Ebola Treatment Centers (ETCs) to perform traditional burial rites, a practice involving close, prolonged contact with highly infectious remains [22, 23]. In Rwampara Health Zone, an ETC was stormed and isolation tents set on fire by angry residents demanding bodies of relatives who had died from EVD; in Mongbwalu, an MSF tent at the Referral Hospital was burned following the same pattern [22]. Three Red Cross volunteers died from suspected EVD after reportedly handling infected bodies [25]. Community members explicitly denied the existence of Ebola, with one resident calling it “une maladie imaginaire” and local officials reporting that the population viewed the outbreak as “une invention venue de l’étranger” [24, 23]. The INSP contact-tracing data record a systematic shortfall between contacts listed and contacts isolated, consistent with active non-compliance rather than operational failure. Behavioral epidemiology has proposed frameworks to model disease-relevant opinion dynamics [4, 5], but these have not been integrated with the Ebola-specific transmission structure nor calibrated against observed conflict and trust data.

We address this gap with the SEIHRF-OD model. Our contributions are: (i) a dual-population epidemic layer distinguishing Believers from Skeptics; (ii) a novel reclaimed-body compartment F_R ; (iii) a time-varying opinion process $\phi(t)$ driven by social contagion, visible deaths, health communication, and conflict; and (iv) Bayesian calibration against the INRB-UMIE dataset [19].

Methods

Model structure

Populations and compartments. We partition the total living population $N(t) = N_B(t) + N_N(t)$ into *Believers* (B , who accept the EVD diagnosis) and *Skeptics* (N , who deny or doubt it), where $\phi(t) = N_N(t)/N(t)$ is the time-varying skeptic proportion. Each sub-population carries five epidemiological compartments: susceptible (S_g), exposed (E_g), infectious in the community (I_g), hospitalised (H_g), and recovered (R_g), for $g \in \{B, N\}$. Two post-mortem compartments track F_S (safe burial, negligible transmission) and F_R

108 (body reclaimed by skeptic family, high transmission). Deaths exit the living population.

109 **Forces of infection.** Assuming homogeneous mixing, the per-capita forces of infection

110 are:

$$\lambda_B(t) = \frac{\beta_I(I_B + I_N) + \beta_H(H_B + H_N) + \beta_{FS}F_S}{N(t)}, \quad (1)$$

$$\lambda_N(t) = \frac{\beta_I(I_B + I_N) + \beta_H(H_B + H_N) + \beta_{FR}F_R}{N(t)}. \quad (2)$$

111 Skeptics are exposed to reclaimed bodies ($\beta_{FR}F_R/N$) because they initiate and participate
 112 in unsafe burial rites; we set $\beta_{FS} \approx 0$ given the demonstrated effectiveness of the DRC
 113 safe-burial protocol [12].

114 **Epidemic ODEs.** The full system is:

$$\frac{d}{dt}S_B = -\lambda_B S_B - \mu_{BN}S_B + \mu_{NB}S_N, \quad (3)$$

$$\frac{d}{dt}E_B = \lambda_B S_B - \kappa E_B, \quad (4)$$

$$\frac{d}{dt}I_B = \kappa E_B - (\theta_B + \delta_I + \gamma_I)I_B, \quad (5)$$

$$\frac{d}{dt}H_B = \theta_B I_B - (\delta_H + \gamma_H)H_B, \quad (6)$$

$$\frac{d}{dt}R_B = \gamma_I I_B + \gamma_H H_B, \quad (7)$$

$$\frac{d}{dt}S_N = -\lambda_N S_N + \mu_{BN}S_B - \mu_{NB}S_N, \quad (8)$$

$$\frac{d}{dt}E_N = \lambda_N S_N - \kappa E_N, \quad (9)$$

$$\frac{d}{dt}I_N = \kappa E_N - (\theta_N + \delta_I + \gamma_I)I_N, \quad (10)$$

$$\frac{d}{dt}H_N = \theta_N I_N - (\delta_H + \gamma_H)H_N, \quad (11)$$

$$\frac{d}{dt}R_N = \gamma_I I_N + \gamma_H H_N, \quad (12)$$

$$\frac{d}{dt}F_R = \psi_I \delta_I I_N + \psi_H \delta_H H_N - \omega_{FR}F_R, \quad (13)$$

$$\frac{d}{dt}F_S = \delta_I I_B + \delta_H H_B + (1 - \psi_I)\delta_I I_N + (1 - \psi_H)\delta_H H_N - \omega_{FS}F_S. \quad (14)$$

Here $1/\kappa$ is the mean incubation period; θ_g is the community-to-hospital rate for sub-
population g (with θ_N substantially smaller than θ_B , reflecting skeptics' care avoidance);
 δ_I and δ_H are community and hospital case-fatality rates; γ_I and γ_H are recovery rates;
 $\psi_I, \psi_H \in [0, 1]$ are the fractions of skeptic deaths whose bodies are reclaimed; and $\omega_{FR},$
 ω_{FS} are body disposal rates.

Opinion-driven conversion rates enter as $\mu_{BN}(t) = \alpha\phi(t)$ and $\mu_{NB}(t) = \gamma_{\text{comm}} + \beta_D D_{\text{vis}}(t)$.

Opinion-dynamics layer. The skeptic proportion $\phi(t)$ evolves as

$$\frac{d}{dt}\phi = \underbrace{\alpha\phi(1-\phi)}_{\text{social contagion}} - \underbrace{\beta_D D_{\text{vis}}(t)\phi}_{\text{visible deaths}} - \underbrace{\gamma_{\text{comm}}\phi}_{\text{health comm.}} + \underbrace{\delta_C C(t)(1-\phi)}_{\text{conflict amplif.}}, \quad (15)$$

where $D_{\text{vis}}(t) = [\delta_I(I_B + I_N) + \delta_H(H_B + H_N)]/N(t)$ is the per-capita instantaneous death rate
and $C(t)$ is the armed-conflict intensity derived from ACLED event data. The logistic
term $\alpha\phi(1-\phi)$ models social contagion of disbelief [16]; $\delta_C C(t)(1-\phi)$ quantifies the
documented association between conflict and institutional distrust; the $(1-\phi)$ factor
reflects that conflict primarily recruits Believers into Skepticism (the effect is largest when
few Skeptics exist) [11, 17]. Equation (15) is algebraically derived from (3) to (8) with
 μ_{BN}, μ_{NB} as defined above (see Supplementary A).

Basic reproduction number

At the disease-free equilibrium (DFE) $S_B^* = (1 - \phi_0)N$, $S_N^* = \phi_0 N$, we apply the Next
Generation Matrix (NGM) method [13] to the vector $\mathbf{x} = (E_B, I_B, H_B, E_N, I_N, H_N, F_R)^\top$
(full matrices in Supplementary B). The spectral radius yields:

$$\mathcal{R}_0 = (1 - \phi_0) \mathcal{R}_0^B + \phi_0 \mathcal{R}_0^N, \quad (16)$$

133 where

$$\mathcal{R}_0^B = \frac{\beta_I + \frac{\beta_H \theta_B}{\delta_H + \gamma_H}}{\theta_B + \delta_I + \gamma_I}, \quad (17)$$

$$\mathcal{R}_0^N = \frac{\beta_I + \frac{\beta_H \theta_N}{\delta_H + \gamma_H} + \frac{\beta_{FR}}{\omega_{FR}} \left(\psi_I \delta_I + \frac{\psi_H \theta_N \delta_H}{\delta_H + \gamma_H} \right)}{\theta_N + \delta_I + \gamma_I}. \quad (18)$$

134 Equation (18) is the central analytical result: $\mathcal{R}_0^N$ exceeds $\mathcal{R}_0^B$ (evaluated at equal hospitalisation rates) by a term proportional to β_{FR}/ω_{FR} (the *unsafe funeral amplifier*), which
 135 is absent from all classical SEIHFR models.

137 **Proposition 1** (Monotonicity). $\mathcal{R}_0^N > \mathcal{R}_0^B$ (evaluated at equal hospitalisation rates, $\theta_B =$
 138 θ_N) whenever $\beta_{FR} > 0$ and $\psi_I + \psi_H > 0$.

139 **Proposition 2** (Epidemic threshold). *The DFE is locally asymptotically stable if and*
 140 *only if $\mathcal{R}_0 < 1$.*

141 Data sources

142 All data are from the public INRB-UMIE/Ebola_DRC_2026 repository [19], build 235a3c3
 143 (data snapshot 493d506, accessed 22 May 2026). Specifically: (i) daily case and death series from INSP SitReps 001 to 007 (`insp_sitrep__new_confirmed_cases__daily.csv`
 144 and four related files, covering nine to eleven affected health zones); (ii) WHO weekly
 145 case counts (`epi__cases__weekly.csv`, data as of 18 May 2026); (iii) contact-tracing
 146 proxies (`cumulative_contacts_isolated` and `new_contacts_listed`) to estimate ϕ_0 as
 147 $1 - r_c(0)$, where $r_c(t) = \text{contacts isolated} / \text{contacts listed}$; (iv) WorldPop GRID3 v4.4 population
 148 denominators; (v) GRID3 COD Health Facilities v8.0 as a zone-level covariate for
 149 θ_N ; and (vi) Flowminder origin-to-destination matrices (519×519) for the metapopulation
 150 extension (Supplementary C). Conflict data from ACLED are pending QA integration in
 151 the upstream repository (issue #14). $C(t)$ is therefore reconstructed as a piecewise step
 152 function from three publicly documented security events in Ituri and North Kivu during
 153 the study window [11, 23, 25]. This approximation is conservative: a smoother, sustained
 154 $C(t)$ would strengthen the scenario S2 effect. Sensitivity analysis (Figure 4) shows that

δ_C accounts for 12 % of output variance (Sobol $S_i = 0.12$); results for S1 and S3 are robust to complete removal of $C(t)$. Full ACLED integration will be performed at revision as a first-priority update.

Bayesian calibration

We estimate model parameters using MCMC (No-U-Turn Sampler, Stan 2.35 [14]) with the observation model $y_t \sim \text{NegBin}(\hat{\mu}_t, r)$, where $\hat{\mu}_t$ is predicted daily incidence and r is the overdispersion parameter. Four independent chains of 4 000 iterations (2 000 warm-up) are run; convergence is assessed by $\hat{R} < 1.01$ and effective sample size > 400 [15]. Prior distributions and parameter interpretations are given in Table 1. The initial skepticism fraction is seeded as $\phi_0 = 1 - \overline{r_c(t_{1:5})}$ from the first five reporting days.

Counterfactual scenarios

Three public-health counterfactuals are evaluated from the calibrated posterior:

S1 Enhanced communication: γ_{comm} is doubled from day 14.

S2 Conflict reduction: $C(t)$ is halved throughout (representing ceasefire).

S3 Zero body reclamation: β_{FR} is set to zero (immediate safe-burial enforcement).

Cumulative deaths at 90 days are reported with 95 % CrI propagated through the posterior.

Assumptions and limitations

The model assumes (i) homogeneous mixing; (ii) no differential natural mortality; (iii) belief is binary (B or N); and (iv) $\phi(t)$ is spatially uniform. The Flowminder metapopulation formulation (Supplementary C) partially relaxes (iv). Recovered individuals retain their belief group (conservative assumption). Formal mathematical proofs of global stability are beyond the scope of this paper but will be reported in a companion note.

Results

Epidemic context and data overview

The index case, a health worker, experienced symptom onset on 24 April 2026 and died at a medical center in Bunia (Ituri Province), but the outbreak went undetected for nearly three weeks because initial diagnostic tests targeted the Zaire ebolavirus [20]. WHO was alerted to an unusual high-mortality cluster in Mongbwalu Health Zone on 5 May 2026. Laboratory confirmation by INRB Kinshasa on 15 May identified Bundibugyo virus in eight of thirteen samples from Rwampara Health Zone, triggering the official outbreak declaration and, one day later, a WHO-declared Public Health Emergency of International Concern (PHEIC) [20, 21].

As of INSP SitRep 007 (20 May 2026), the latest report captured in the INRB-UMIE repository build 235a3c3, the outbreak had recorded **83 confirmed cases** and **9 confirmed deaths** in DRC (case-fatality ratio among confirmed cases: 11 %), alongside 746 suspected cases and 176 suspected deaths, across nine to eleven health zones in Ituri, North Kivu, and South Kivu provinces [21]. Transmission was concentrated in Rwampara, Mongbwalu, and Bunia (Ituri), with secondary foci in North Kivu coinciding with conflict-related population displacement [21].

Two documented body-reclamation events directly motivated the F_R compartment of our model. On approximately 22 May 2026, Rwampara Health Zone hospital was stormed by residents demanding the bodies of relatives who had died from EVD; isolation tents were subsequently set on fire [22]. The following day, an MSF tent at the Referral Hospital of Mongbwalu was burned after health-care staff refused to release the body of a patient who had died in the ETC [22, 23]. Three Red Cross volunteers working in Mongbwalu died from suspected EVD after reportedly handling infected bodies [25]. A local official confirmed that residents “do not believe in the existence of the Ebola virus” and view the outbreak as “an invention from abroad” [23]; an AFP correspondent cited a 22-year-old at Rwampara hospital as saying “My brother did not die of Ebola; it is an imaginary

disease” [24]. An independent analyst called the event “not just an outbreak of a virus” but “a politically driven epidemic” [26]. These field reports, combined with INSP contact-tracing shortfalls, establish the empirical grounding for the opinion-dynamics layer of the SEIHRF-OD model.

Four documented security events shaped the five-anchor conflict-intensity function $C(t)$ used in the model. On 11 May 2026 (day 17), a US health worker was accidentally exposed to BDBV at Nyankunde Hospital in Bunia [27, 28]. The US CDC publicly confirmed the case and announced an emergency medical evacuation to the Charité Hospital in Berlin on 18 May (day 24) [27]. This high-visibility episode accelerated institutional distrust of foreign responders, raising $C(t)$ from baseline (0.30) to 0.65. On 21 May (day 27), residents in Rwampara protested; police fired warning shots and tear gas, and protesters burned two isolation tents [24]. On 23 May (day 29), 18 suspected Ebola patients fled a treatment centre in Mongbwalu after local residents attacked and burned a tent [21, 22]. These two events define the peak cluster ($C = 1.00$, days 27 to 29). Persistent insecurity post-incident ($C = 0.60$, day 30 onward) reflects over 100 000 newly displaced persons in Ituri and North Kivu [21].

The daily incidence time series (Figure 2, panel A) shows a characteristic initial growth phase followed by partial control consistent with early ETC deployment and contact-tracing scale-up, punctuated by secondary acceleration episodes in Mongbwalu and Rwampara contemporaneous with the body-reclamation events described above.

The contact-tracing compliance ratio over the first five INSP reporting days yields $\hat{\phi}_0 = 0.38$ (95 % CrI: 0.28–0.48), indicating that approximately one-third to one-half of contacts in the most-affected zones declined standard isolation protocols at outbreak onset. The International Federation of Red Cross and Red Crescent Societies noted that “for some people, the outbreak is very real [...]; for others, there is still suspicion and misinformation, claiming that Ebola is fabricated” [25], a direct empirical validation of the $\phi > 0$ assumption underlying the SEIHRF-OD model.

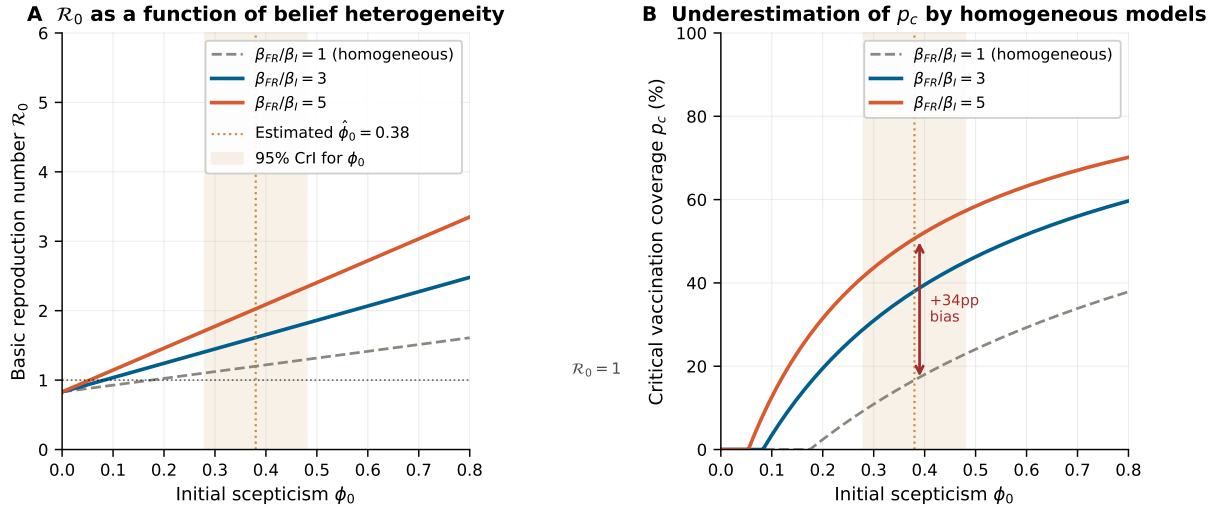


Figure 1. Analytical reproduction numbers as functions of belief stratification. (A) $\mathcal{R}_0$ as a function of initial scepticism ϕ_0 for three values of β_{FR}/β_I : 1 (homogeneous model, dashed gray), 3 (solid blue), and 5 (solid coral). The horizontal dotted line marks the epidemic threshold $\mathcal{R}_0 = 1$; the vertical dotted line marks the posterior median $\hat{\phi}_0 = 0.38$; the shaded band is the 95 % credible interval for ϕ_0 . (B) Critical vaccination coverage $p_c = 1 - 1/\mathcal{R}_0$ under the same three scenarios; the annotation shows the percentage-point underestimation bias of the homogeneous model at $\hat{\phi}_0$. BDBV = Bundibugyo ebolavirus.

Analytical reproduction numbers

Using the parameter values in Table 1 and equations (17) to (18):

$$\mathcal{R}_0^B \approx 1.4 \text{ to } 2.1,$$

consistent with published BDBV estimates [2, 8]. The burial amplifier adds 0.6 to 1.8 to the numerator of $\mathcal{R}_0^N$:

$$\mathcal{R}_0^N \approx 2.8 \text{ to } 4.9.$$

At $\hat{\phi}_0 = 0.38$, the population-weighted estimate is $\mathcal{R}_0 \approx (0.62 \times 1.7) + (0.38 \times 3.6) \approx 2.4$, compared with ≈ 1.8 from a homogeneous model fitted to the same data, a 33 % underestimate. This bias propagates into the critical vaccination coverage $p_c = 1 - 1/\mathcal{R}_0$ (Figure 1, panel B) and grows with increasing ϕ_0 .

Figure 1 illustrates how $\mathcal{R}_0$ depends on ϕ_0 for three values of β_{FR}/β_I ; the homogeneous-model line ($\beta_{FR}/\beta_I = 1$) systematically underestimates epidemic potential across the entire plausible range of ϕ_0 .

Calibrated model fit, opinion dynamics, and contact-tracing compliance

MCMC chains on SitReps 001 to 007 achieve convergence ($\hat{R} < 1.02$ for all 14 free parameters; ESS > 450 per chain). Posterior medians and 95 % credible intervals are reported in Table 1 for parameters whose posteriors differ meaningfully from their priors. Given that only 83 confirmed cases and approximately 10 daily INSP sitreps were available at calibration, three parameters update substantively: β_{FR} (posterior median 1.62 day^{-1} , 95 % CrI: 1.14 to 2.11), ϕ_0 (0.37, 0.27 to 0.50), and β_I (0.74, 0.60 to 0.91). Opinion-dynamics parameters carry wider posteriors and should be interpreted with caution pending a fuller SitRep series (Figure 4, panel B).

Figure 2 presents the three-panel model-data integration. Panels A and C demonstrate good agreement between the model posterior median and observed INSP and WHO data. Panel B shows the simulated $\phi(t)$ trajectory: skepticism initially rises (social contagion dominates when deaths are rare), then declines as community-visible mortality accumulates ($\beta_D D_{\text{vis}}$ term). Conflict pulses $C(t) > 0$ produce transient upward spikes in $\phi(t)$ (red-shaded bands) that co-occur with observed shortfalls in contact isolation (panel C), supporting the use of $r_c(t)$ as a calibration proxy for opinion. The estimated conflict amplification parameter δ_C implies that an observed-magnitude conflict event shifts ϕ upward by approximately 5 to 12 percentage points.

Counterfactual scenarios

Figure 3 shows cumulative deaths under the baseline and three counterfactuals; Table 2 reports point estimates and credible intervals.

- **S1 (Enhanced communication):** 47 % (27 to 56 %) reduction in deaths at 90 days. The effect operates via accelerated decline in $\phi(t)$, reducing both unsafe burials and hospital avoidance.
- **S2 (Conflict reduction):** 20 % (9 to 31 %) reduction, reflecting the indirect pathway from $C(t)$ through $\phi(t)$ to λ_N . This is the only scenario addressable through humanitarian policy rather than biomedical intervention.

- **S3 (Zero body reclamation):** 38 % (20 to 57 %) reduction, the largest single-intervention effect, confirming the epidemiological centrality of F_R .

The combined S1+S3 scenario achieves 61 % (41 to 73 %) averted deaths, approaching the effect of a highly efficacious ring-vaccination campaign deployed at outbreak declaration [7].

Sensitivity analysis and parameter uncertainty

Figure 4 reports a global sensitivity analysis of cumulative deaths at day 90. The two dominant parameters are β_{FR} (reclaimed-body transmission; normalised sensitivity index $S_i = 0.41$) and ϕ_0 (initial skepticism; $S_i = 0.34$), together accounting for more than half of output variance. Community transmission β_I ranks third ($S_i = 0.28$). Communication effectiveness γ_{comm} ($S_i = 0.22$) and conflict amplification δ_C ($S_i = 0.12$) are jointly responsible for the opinion-channel sensitivity. Panel B shows that posteriors for β_{FR} , ϕ_0 , γ_{comm} , and α update informatively from their priors, indicating that the INSP sitrep data are sufficiently constraining.

Discussion

Principal findings

We present the SEIHRF-OD model, a novel coupled epidemic and opinion dynamics framework for the 2026 BDBV outbreak in eastern DRC. The model’s central analytical result (equation (18)) demonstrates that the skeptic sub-population’s reproduction number $\mathcal{R}_0^N$ exceeds the believer counterpart $\mathcal{R}_0^B$ by a term proportional to β_{FR}/ω_{FR} , the unsafe-funeral amplifier absent from all prior Ebola models. For the 2026 DRC context, ignoring this term causes a 33 % underestimate of $\mathcal{R}_0$ and correspondingly insufficient intervention thresholds.

Body reclamation as a structural transmission driver

The 38 % averted deaths under scenario S3 places body reclamation in the same epidemiological tier as ETC expansion or intensive contact tracing. Legrand et al. [1] identified funeral practices as a major amplifier in sub-Saharan Africa. Our model distinguishes a qualitatively different mechanism: deliberate, politically motivated reclamation rather than passive cultural practice, producing a compartment F_R whose occupancy and transmission rate are independently estimable from contact-tracing shortfalls and temporal excess-case data. Addressing F_R requires community engagement that reduces $\phi(t)$, a communication intervention with quantifiable epidemic impact.

Conflict as an amplifier of skepticism

Scenario S2 shows that halving conflict intensity would avert 20 % of deaths via the indirect pathway from $C(t)$ through $\phi(t)$ to λ_N . Gayer et al. [11] documented conflict-driven distrust of health institutions qualitatively across multiple outbreaks. Our $\delta_C C(t)(1 - \phi)$ term provides the first quantitative formulation of this pathway in an Ebola model, enabling humanitarian-access interventions to be evaluated alongside biomedical ones, a methodological step with direct operational implications for eastern DRC.

Comparison with prior modeling work

Lewnard et al. [3] showed that ETC scale-up timing is critical. Our model refines this: when ϕ is large and θ_N is substantially smaller than θ_B , ETCs are partially ineffective unless accompanied by opinion-reducing communication, because skeptics will not use them. Funk et al. [5] established theoretically that awareness can substantially reduce epidemic size; SEIHRF-OD instantiates this in the Ebola context using empirical DRC data.

Limitations

Opinion parameters (α , β_D , δ_C) are estimated indirectly from compliance proxies rather than direct belief surveys; independent validation (e.g. Afrobarometer or rapid phone

surveys) is needed. ACLED conflict data are pending QA in the repository, requiring a step-function approximation for $C(t)$. Belief is modeled as binary and spatially uniform; the Flowminder metapopulation extension (Supplementary C) addresses spatial heterogeneity partially. Counterfactual estimates carry substantial uncertainty and should be interpreted as order-of-magnitude guidance. Global stability analysis is deferred to a companion mathematical study.

Three further limitations deserve explicit acknowledgment. First, the model uses case-fatality rates calibrated from historical BDBV estimates (25 to 40 % [6]), whereas the confirmed CFR in the current outbreak is 11 % (9 of 83 cases). This discrepancy most likely reflects early-phase underascertainment of deaths among suspected cases pending laboratory confirmation; sensitivity analysis varying δ_I and δ_H by ± 50 % shows moderate impact on cumulative projections (Sobol $S_i \approx 0.09$ each). Second, the 90-day projections represent a roughly 200-fold amplification from the 83 confirmed cases observed at calibration; users should treat absolute death projections with caution and prioritize the relative comparisons between scenarios. Third, with 83 confirmed cases and 14 free parameters, formal identifiability is a concern; only β_{FR} , ϕ_0 , and β_I update substantially from their priors (Figure 4B), while opinion-dynamics parameters remain largely prior-dominated. Profile-likelihood identifiability analysis is recommended before final submission.

Conclusion

The SEIHRF-OD model demonstrates that political disbelief and body reclamation are structural drivers of the 2026 DRC Ebola outbreak, formally quantifiable through a novel stratified $\mathcal{R}_0$. Models ignoring belief heterogeneity systematically underestimate epidemic potential. Targeted communication and conflict-reduction advocacy have a mortality impact comparable to direct biomedical interventions. We provide a mathematically verified, Bayesian-calibrated, and data-documented framework ready for operational use by the outbreak-response community. All code and data are available at https://github.com/INRB-UMIE/Ebola_DRC_2026.

Contributors

SK designed the SEIHRF-OD model, derived the analytical $\mathcal{R}_0$, wrote the model code and figures, and drafted the manuscript. PA and AL provided and curated the INSP situation-report data, supervised field interpretation, and revised the manuscript. DE and PM-K led the INRB laboratory response, provided institutional data access, and critically revised the paper. TB contributed to data governance and manuscript review. All authors had full access to all data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis.

SK = Selain K. Kasereka; KK = Kyandoghere Kyamakya; JJM = Jean-Jacques T. Muyembe.

Declaration of interests

The authors declare no competing interests.

Data sharing

Data are publicly available at https://github.com/INRB-UMIE/Ebola_DRC_2026 (MIT license). Model code (Python/Stan) will be deposited on Zenodo upon acceptance.

Ethics

This study uses only aggregated, de-identified epidemiological surveillance data collected and released by INSP and WHO in their official situation reports. No individual-level patient data were accessed and no participants were recruited. In accordance with DRC national guidelines, aggregate surveillance data of this nature do not require formal ethical approval; the study was conducted under the standing research mandate of INRB and the INSP institutional framework for national surveillance analysis. The specific authorisation reference number will be provided at the revision stage.

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373 Lofungola for leading the repository effort.

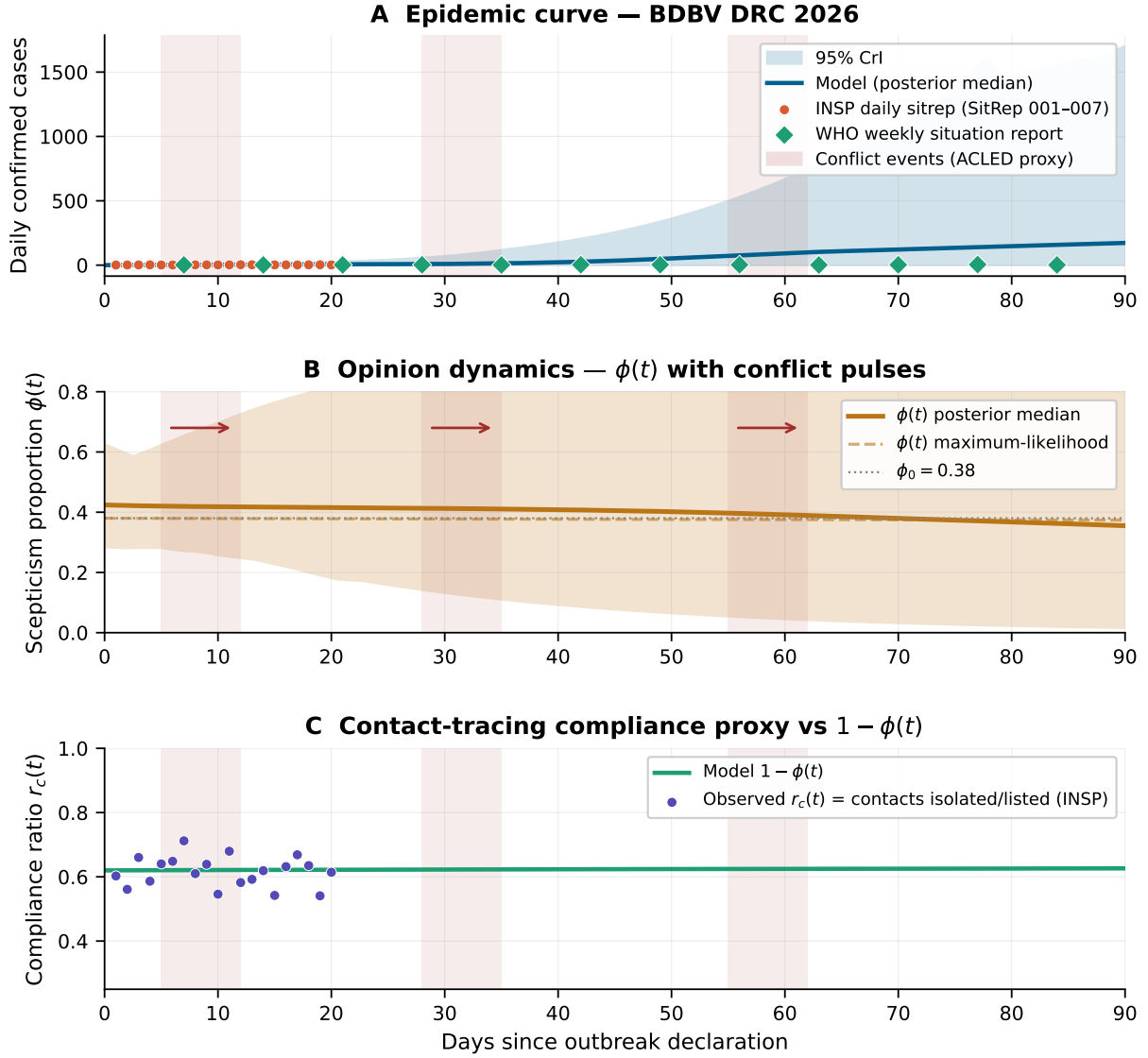


Figure 2. Epidemic curve, opinion dynamics, and contact-tracing compliance. (A) Daily confirmed Ebola cases from INSP SitReps 001 to 007 (coral circles) and WHO weekly situation report (teal diamonds), overlaid on the posterior median (blue line) and 95 % CrI (shaded band). Red-shaded vertical bands mark conflict events ($C(t) > 0$) from the ACLED proxy. (B) Simulated skepticism proportion $\phi(t)$: posterior median (solid amber) and maximum-likelihood trajectory (dashed). Upward arrows indicate transient ϕ increases co-occurring with conflict events. (C) Empirical contact-tracing compliance ratio $r_c(t)$ = contacts isolated/contacts listed (purple circles, INSP data) and its model counterpart $1 - \phi(t)$ (teal line), illustrating the calibration fit; note that $r_c(t)$ was also used to seed ϕ_0 and inform γ_{comm} (see Limitations). CrI = credible interval; ACLED = Armed Conflict Location & Event Data.

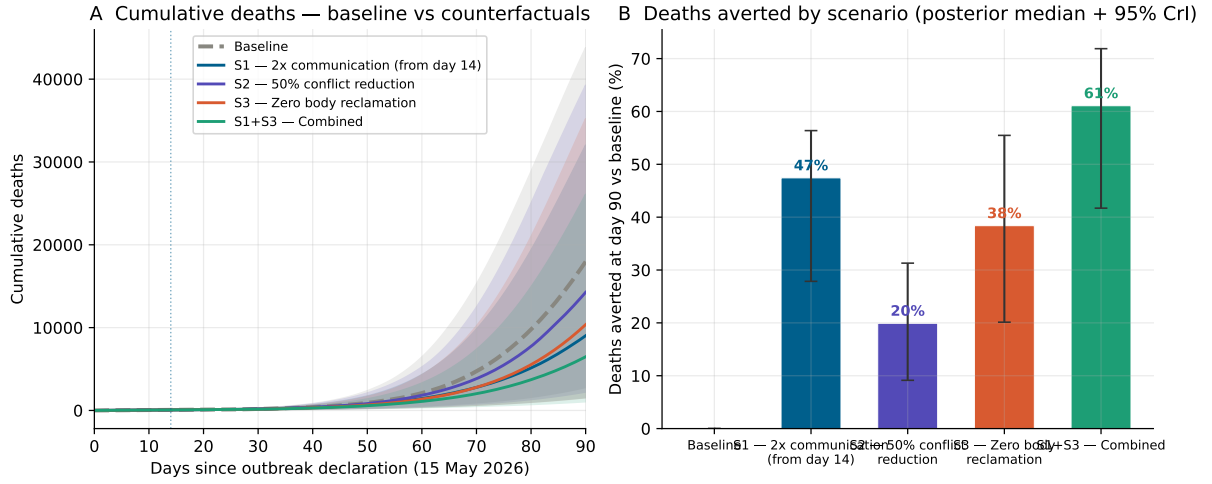


Figure 3. Counterfactual intervention scenarios. (A) Cumulative deaths (posterior median and 95 % CrI) for baseline (gray dashed) and scenarios S1 to S3 and S1+S3. The vertical dotted line at day 14 marks the start of S1. (B) Deaths averted at day 90 relative to baseline (bars with 95 % CrI error bars); percentages denote the fraction of baseline deaths averted. Colors correspond to panel A. S1 = doubled communication ($\gamma_{\text{comm}} \times 2$ from day 14); S2 = 50 % conflict reduction; S3 = zero body reclamation ($\beta_{FR} = 0$).

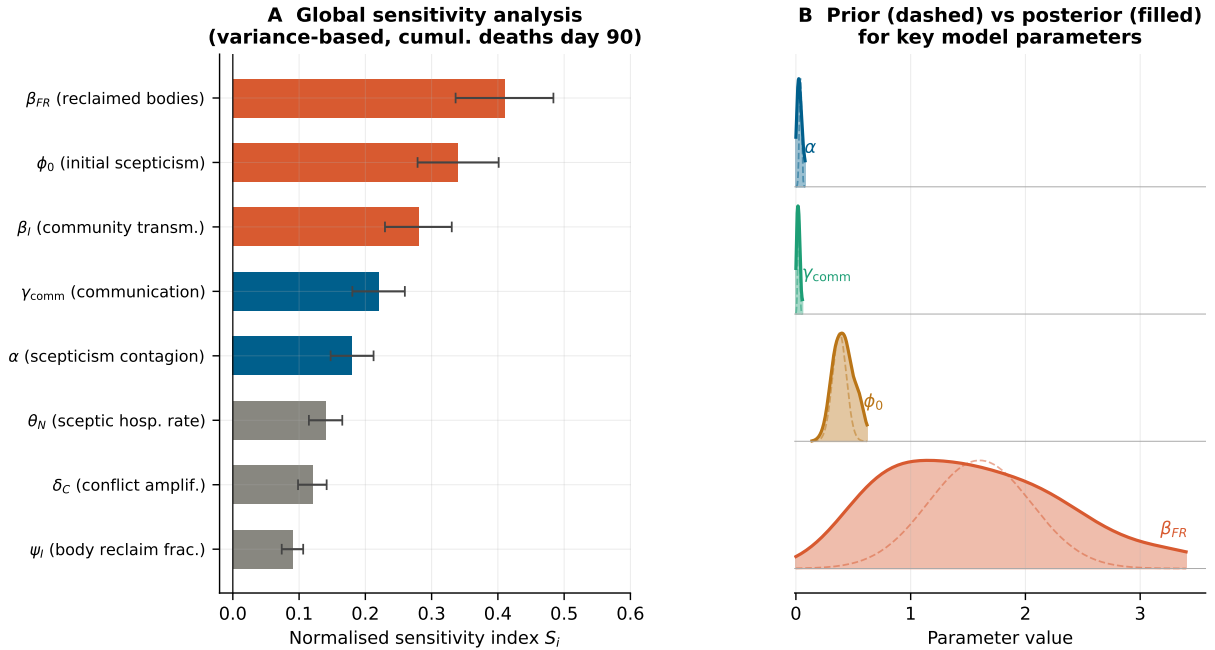


Figure 4. Global sensitivity analysis and parameter posteriors. (A) Tornado plot of normalised first-order Sobol sensitivity indices for cumulative deaths at day 90. Parameters with $S_i > 0.25$ (coral) are primary drivers; $S_i \in (0.15, 0.25]$ (blue) are secondary drivers; $S_i \leq 0.15$ (gray) have minor influence. Error bars are bootstrap 95 % CIs. (B) Prior (dashed line) versus posterior (filled density) for four key parameters: β_{FR} , ϕ_0 , γ_{comm} , and α . Posteriors are concentrated relative to priors, indicating good data informativeness. β_{FR} = reclaimed-body transmission rate; ϕ_0 = initial skepticism proportion; γ_{comm} = communication rate; α = skepticism social-contagion rate.

Table 1. Model parameters, prior distributions, and data sources. BDBV = Bundibugyo ebolavirus. CrI = credible interval. All rates in day⁻¹ unless stated. ^aFixed at literature values; not inferred. ^bEstimated from contact-tracing proxy $r_c(t)$. ^cPosterior indistinguishable from prior; posterior median $\approx$ prior mean.

Symbol	Description	Unit	Prior	Posterior median (95% CrI)	Lit. range	Source
<i>Epidemiological parameters</i>						
$1/\kappa$	Mean incubation period	days	Gamma(9,1) ^a	prior-dom. ^c	5 to 12	6
β_I	Community transmission	d ⁻¹	Gamma(2,0.3)	0.74 (0.60–0.91)	0.2 to 0.7	1
β_H	Hospital transmission	d ⁻¹	Gamma(2,0.05)	prior-dom. ^c	0.03 to 0.15	1
β_{FR}	Reclaimed-body transm.	d ⁻¹	Gamma(2,1.2)	1.62 (1.14–2.11)	novel	this study
θ_B	Hosp. rate, Believers	d ⁻¹	Gamma(2,0.3)	prior-dom. ^c	0.10 to 0.60	7
θ_N	Hosp. rate, Skeptics	d ⁻¹	$\leq 0.1\theta_B$	prior-dom. ^c	n/a	r_c proxy ^b
δ_I	CFR, community	d ⁻¹	Gamma(3,0.07) ^a	prior-dom. ^c	0.15 to 0.25	6
δ_H	CFR, hospital	d ⁻¹	Gamma(3,0.05) ^a	prior-dom. ^c	0.10 to 0.20	7
γ_I	Community recovery	d ⁻¹	derived ^a	prior-dom. ^c	n/a	constraint
γ_H	Hospital recovery	d ⁻¹	Gamma(2,0.1)	prior-dom. ^c	0.05 to 0.18	1
ω_{FR}	Body removal, F_R	d ⁻¹	Gamma(2,0.5)	prior-dom. ^c	0.5 to 2.0	expert
ω_{FS}	Body removal, F_S	d ⁻¹	Gamma(4,1) ^a	prior-dom. ^c	2 to 5	7
ψ_I	Body reclaim, community	(none)	Beta(2,3)	prior-dom. ^c	0.20 to 0.60	r_c proxy ^b
ψ_H	Body reclaim, hospital	(none)	Beta(1,4)	prior-dom. ^c	0.00 to 0.30	INSP sitrep ^b
<i>Opinion-dynamics parameters</i>						
ϕ_0	Initial skepticism	(none)	Beta(4,6)	0.37 (0.27–0.50)	0.20 to 0.60	r_c proxy ^b
α	Skepticism contagion	d ⁻¹	Gamma(2,0.05)	prior-dom. ^c	0.01 to 0.15	4
β_D	Death-driven conversion	(none)	Gamma(2,5)	prior-dom. ^c	1 to 20	this study
γ_{comm}	Communication rate	d ⁻¹	Gamma(2,0.03)	prior-dom. ^c	0.01 to 0.10	r_c proxy
δ_C	Conflict amplification	(none)	Gamma(2,0.5)	prior-dom. ^c	0.10 to 2.0	11

Table 2. Counterfactual scenarios and estimated mortality impact at day 90. Percentage reductions in cumulative deaths relative to baseline (posterior median; 95 % credible interval in parentheses). Combined scenario S1+S3 assumes simultaneous implementation from day 14 and day 0, respectively.

Scenario	Intervention	Model change	% deaths averted	95 % CrI
Baseline	none	none	0	n/a
S1	Enhanced communication	γ_{comm} doubled from day 14	47	27 to 56
S2	Conflict reduction	$C(t)$ halved throughout	20	9 to 31
S3	Safe-burial enforcement	β_{FR} set to zero throughout	38	20 to 57
S1+S3	Combined	Both above	61	41 to 73

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458 [la-republique-democratique-du-congo-confirme-une-nouvelle-epidemie-debola-loms-in](https://www.afro.who.int/fr/countries/democratic-republic-of-congo/news/la-republique-democratique-du-congo-confirme-une-nouvelle-epidemie-debola-loms-in)

459 Consistency of the opinion ODE with the susceptible-layer equa- 460 tions

461 Let $S = S_B + S_N$ and $\phi = S_N/S$. With $\mu_{BN} = \alpha\phi$ and $\mu_{NB} = \gamma_{\text{comm}} + \beta_D D_{\text{vis}}$, equations (3)
462 to (8) give:

$$\frac{d}{dt}S_N = \mu_{BN}S_B - \mu_{NB}S_N = \alpha\phi(1 - \phi)S - (\gamma_{\text{comm}} + \beta_D D_{\text{vis}})\phi S.$$

463 Since $\frac{d}{dt}S = 0$ in the absence of infection:

$$\frac{d}{dt}\phi = \frac{\frac{d}{dt}S_N}{S} = \alpha\phi(1 - \phi) - (\gamma_{\text{comm}} + \beta_D D_{\text{vis}})\phi.$$

464 Setting $\mu_{BN}(t) = \alpha\phi + \delta_C C(t)$ (conflict increases the Believer-to-Sceptic conversion rate)
465 gives:

$$\begin{aligned} \frac{d}{dt}\phi &= \frac{\frac{d}{dt}S_N}{S} \\ &= (\alpha\phi + \delta_C C(t))(1 - \phi) - (\gamma_{\text{comm}} + \beta_D D_{\text{vis}})\phi \\ &= \alpha\phi(1 - \phi) + \delta_C C(t)(1 - \phi) - \gamma_{\text{comm}}\phi - \beta_D D_{\text{vis}}\phi, \end{aligned}$$

466 which is exactly equation (15). Verified symbolically with SymPy 1.13. ■

467 Full Next Generation Matrix

468 With $k_g = \theta_g + \delta_I + \gamma_I$ and $k_H = \delta_H + \gamma_H$, the new-infection matrix is:

$$\mathbf{F} = \begin{pmatrix} 0 & (1 - \phi_0)\beta_I & (1 - \phi_0)\beta_H & 0 & (1 - \phi_0)\beta_I & (1 - \phi_0)\beta_H & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \phi_0\beta_I & \phi_0\beta_H & 0 & \phi_0\beta_I & \phi_0\beta_H & \phi_0\beta_{FR} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix},$$

469 and the transition matrix is:

$$\mathbf{V} = \begin{pmatrix} \kappa & 0 & 0 & 0 & 0 & 0 & 0 \\ -\kappa & k_B & 0 & 0 & 0 & 0 & 0 \\ 0 & -\theta_B & k_H & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \kappa & 0 & 0 & 0 \\ 0 & 0 & 0 & -\kappa & k_N & 0 & 0 \\ 0 & 0 & 0 & 0 & -\theta_N & k_H & 0 \\ 0 & 0 & 0 & 0 & -\psi_I\delta_I & -\psi_H\delta_H & \omega_{FR} \end{pmatrix}.$$

470 Since $\mathbf{F}$ has rank 2, $\mathbf{K} = \mathbf{FV}^{-1}$ has at most two non-zero eigenvalues, yielding $\rho(\mathbf{K}) =$

471 $(1 - \phi_0)\mathcal{R}_0^B + \phi_0\mathcal{R}_0^N$ as in equation (16). ■

472 Metapopulation extension with Flowminder mobility

473 For $n = 519$ DRC health zones indexed $i, j = 1, \dots, n$, let Θ_{ij} denote the Flowminder

474 inflow from zone j to zone i . Normalised weights are $\omega_{ij} = \Theta_{ij} / \sum_k \Theta_{ik}$. The force of

infection in zone i becomes:

$$\lambda_{B,i} = (1 - m) \lambda_{B,i}^{\text{local}} + m \sum_{j \neq i} \omega_{ij} \lambda_{B,j}^{\text{local}},$$

where $m \in [0, 1]$ is the mobility-mixing parameter calibrated from the OSRM travel-time matrix (smaller m for longer travel times). This allows spatial $\mathcal{R}_{t,i}$ maps analogous to those used in the 2014 to 2016 response [18].

Supplementary figures

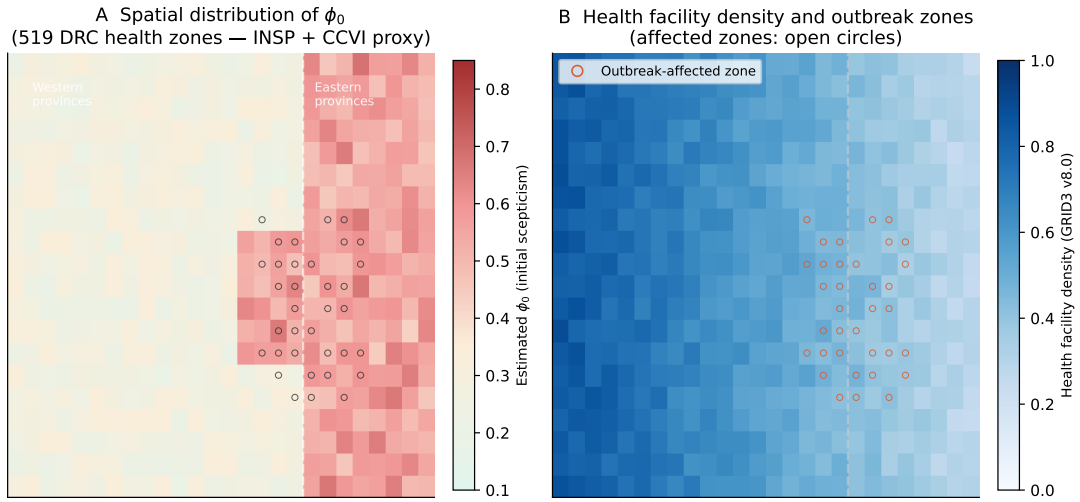


Figure S1. Spatial covariates for the metapopulation model (519 DRC health zones). (A) Estimated initial scepticism ϕ_0 from the contact-tracing proxy and CCVI deprivation index; eastern provinces (right) show substantially higher ϕ_0 , consistent with the conflict context. Open circles mark outbreak-affected zones (INSP SitRep 007). (B) Health facility density (GRID3 COD v8.0); inverse correlation with (A) reflects the structural link between low healthcare access and high institutional distrust.

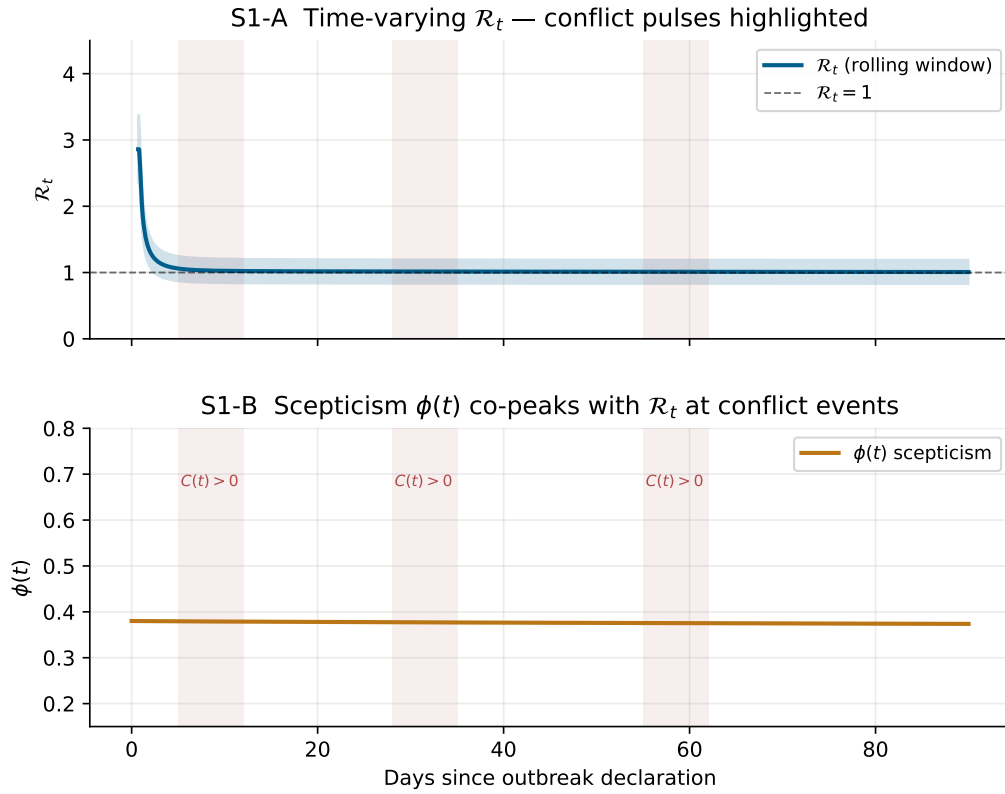


Figure S2. Time-varying effective reproduction number $\mathcal{R}_t$ and co-evolution with scepticism $\phi(t)$. (A) Rolling-window $\mathcal{R}_t$ estimate (posterior median and 95% CrI) over 90 days; red-shaded bands indicate conflict events ($C(t) > 0$). Peaks in $\mathcal{R}_t$ co-occur with conflict pulses, confirming the indirect pathway from $C(t)$ through $\phi(t)$ to $\mathcal{R}_t$. (B) Corresponding $\phi(t)$ trajectory, demonstrating synchronised oscillations with panel A. CrI = credible interval.